United International **University** (**UIU**)

Dept. of Computer Science and Engineering (CSE)

Mid Term Exam Year: **2025** Semester: **Summer**

Course: **CSE 3711/EEE 4413** Title: **Computer Networks (Section – ALL)**

Marks: **30** Time: **1 Hour 30 Minutes**

[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

- There are **3 (Three)** questions. Answer **<u>all 03 questions</u>**. All questions are of values indicated on the right-hand margin.

Q1.

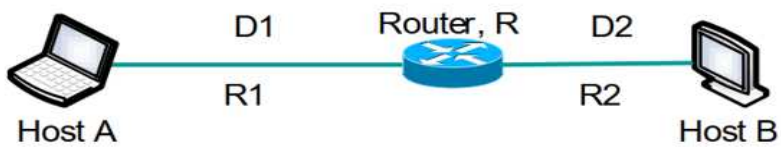
a) Suppose **Alice** and **Bob** are communicating over a TCP/IP network, with a **router (T)** and a **link-layer switch (S)** positioned between the two hosts. Each of these devices (hosts, router and switch) implements a **different set of TCP/IP layers**, reflecting their distinct functionalities.

I. With the aid of **diagrams**, demonstrate the **encapsulation and de-encapsulation** processes. **[3]**

II. Among the following protocols **IP, UDP, Ethernet, SMTP, TCP, and HTTP**, which protocols are implemented in the **intermediate nodes** and **why**. **[1]**

III. When **Alice** and **Bob** communicate **via email**, to **identify** the receiving process, **how many pieces of information** need to be specified and what are they? Illustrate with **an example**. **[2]**

b) Suppose, **Host A** is sending **a message** of length **500 KB** to **Host B**. Bandwidths of two segments are **R1= 5 Mbps** and **R2 = 1 Mbps**. Distances of two segments are **D1 = 2 km** and **D2 = 1 km** respectively. If **$2x10^8$ m/s** is the **propagation speed** in medium, find out the **end-to-end delay** from Host A to Host B while **processing and queuing delays** inside the router is **5 msec** and **10 msec** respectively. **[3]**



c) Suppose that Alice and Bob share a **1 Mbps link**, where every user in that link reserves **10 Kbps**, then in **circuit switching** technology **how many users** may be served if each user is **active 10%** of the time? **Justify** your answer. **[2]**

d) Suppose that ***R_s*** **bits/sec** is the **link capacity** between **Alice** and **S**, whereas ***R_T*** **bits/sec** is the capacity between **Bob** and **T**. If the value of R_T is **2 Mbps** and the value of R_S is **5 Mbps**. What is the **throughput**? **[1]**

Q2.

a) **HTTP** and **SMTP** are both file transfer protocols and have many common characters; for example, they both are client-server protocols and run on top of TCP. List **any two differences** between HTTP and SMTP? **[2]**

b) Suppose that a ***Web server*** and an ***FTP server*** run on **port number 80** and **21** respectively in ***Host C***, which is currently receiving requests from **two different Hosts**, **A** and **B**. Browsers of **Host A** run on **port number 5555** and **Host B** is requesting FTP service using **port number 6666**. With the help of a **diagram** explain **how** does Host C directs segments to the **appropriate socket**? What **tuples (IP: Port)** are being used here? **[3]**

c) I. Explain how ***Symmetric key cryptography*** and ***Public Key cryptography*** work and what are their advantages & disadvantages. **[2]**

II. What are the differences between Message ***Confidentiality*** and message ***Integrity***? **[1]**

III. Can you have confidentiality without integrity or can you have integrity without confidentiality? Justify your answer. **[1]**

Q3.

a) Consider a client accessing a **base HTML page** with **four (4) URL references** to images to be displayed in the page. Assume that it takes **time T1** to establish a TCP connection, **time T2** to retrieve the base page, and **time T3** to retrieve each of the images. **Ignore** time to close a TCP connection, and time to send HTTP request packets. **[3]**

I. **Calculate** the **total time (in terms of T1, T2 and T3)** it takes to retrieve the base page and its embedded images for **non-persistent and persistent HTTP**.

II. **Draw** the corresponding **sequence diagram (steps)** for the above scenario.

b) Suppose **Rahman**, with a Web-based e-mail account (rahman@gmail.com), sends an email to **Ali** (ali@hotmail.com), who accesses his email using **IMAP**. With a **diagram explain** how the email moves from **Rahman's host** to **Ali's mailbox**. Be sure to **identify the application-layer protocols** that are used to move the email between the two hosts. **[3]**

c) Consider a scenario with **Host A**, **local DNS server B**, and the **rest of the DNS infrastructure**. Host A asks Host B to resolve ***www.eLMS.uiu.ac.bd***. **List the steps** taken to resolve ***www.eLMS.uiu.ac.bd*** *and* respond to A. In each step **identify who** B contacts. **[3]**

End of Paper – Thank You